SPACE INDUSTRIALIZATION: SOME LEGAL AND POLICY CONSIDERATIONS FOR PRIVATE ENTERPRISE

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Introduction

In little more than twenty years mankind has experienced unprecedented transition from the confines of earth to the unlimited opportunities offered by routine access to surrounding space. With the launching of the Space Shuttle a new era will be initiated in the saga of space. The reusable Space Shuttle is being developed by the National Aeronautics and Space Administration to deliver payloads to earth orbit more economically than is possible with existing expendable launch vehicles. The addition of the manned laboratory, Spacelab, under development by the European Space Agency, as well as other elements of the new space transportation system, will provide still further capabilities. Engineers, technicians and scientists will be able to remain in space for extended periods of time with substantial operational and mission flexibility and return to earth with the products of their efforts or leave in space, to be revisited as necessary, facilities of continuing usefulness.

During the short period of mankind's efforts in space, considerable change has been occurring in the objectives and in the content of space programs of the United States as well as those of other nations. Exploration for basic human knowledge, technical challenge, and national prestige were the major thrusts and rewards, and dictated a dominant role for government. Application programs derived from scientific and technical advances are adding new dimensions to space activities and are now offering new and fascinating uses of space for direct benefit on earth. Currently, space systems for communications, weather forecasting and earth resources monitoring and detection are providing early evidence of services available from space. Experiments on Apollo and Skylab manned space missions have demonstrated new materials and materials processing capabilities not available on earth, which hold great promise for use in making products with new and unique characteristics.

Role of the Private Sector

The National Aeronautics and Space Act of 1958,[1] mandating that "activities in space should be devoted to peaceful purposes for the benefit of all mankind"[2] galvanized the national effort in space. Carried out under the aegis of the National Aeronautics and Space Administration, the private sector of the country provided the necessary technical support and industrial capabilities to serve the government

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[1] 42 USC 2451 (1958).

[2] 42 USC 2451(a) (1958).

customer. The cooperation of the public and private sectors provided maximum coordination in expanding research and development capabilities to accomplish the goals of the national space program.

The prospects for commercial uses of space for goods and services, however, will provide a new and greater role for the private sector. While NASA responsibilities in government sponsored programs will continue to advance national efforts in space exploration as well as in developing necessary operational capabilities for future requirements, opportunities for broader participation of the private sector are apparent. Activities can be initiated with objectives that are quite separate and independent from those which are directed by NASA and other government agencies.

Characteristics of the space environment such as global viewing, worldwide transmission of information, microgravity, vacuum and abundant solar energy are available for all to use as human intellect and imagination determine. Many of the concepts being investigated by NASA as well as industry which were presented to the Subcommittee on Space Science and Applications during hearings indicate that present uses of space in many respects can be considered quite rudimentary.[3]

Prospects for further improvements in present services as well as energy production, manufacturing, and industries which in all probability have not even been contemplated yet will move the United States as well as other nations to expanded uses of the space environment that will reap countless economic and societal benefits. We are on the threshold of industrializing space and the role of private enterprise will determine the success of these endeavors.

National Interests in Commercialization

Private enterprise interest in space will center around goods and services for use on earth which can be produced either uniquely in space or better in space than on earth. Private sector innovation and creativity in product development and services as well as understanding of marketplace needs dictate the importance of participation from broad and diverse industries and businesses. Greater potential for making the benefits of space available to all peoples of the world can best be realized with the active participation of the private sector. From the viewpoint of the United States, industries, businesses and individuals are where the nation must look for economic strength and this will be no less true in exploiting space and establishing national capabilities in space industrialization.

The importance of private enterprise in the industrialization process will challenge the nation's ability to accommodate the requirements of the free enterprise system in

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[3]U. S. Congress, House Committee on Science and Technology. Space Industrialization, Heatings Before the Subcommittee on Space Science and Applications, September 29, 1977, 95th Cong., 1st Sess., at 204 (1977).

the national economic interest. In testimony before the Committee on Science and Technology, the Comptroller General of the United States gave some provoking advice. Elmer Staats advised that:

> "One of the important areas for the Committee's consideration is how we can maximize private sector research and development which has concentrated in recent years on low-risk, short-term projects directed principally at improving existing products. Emphasis on long-term projects that could lead to new products and processes has decreased. For example, industry now spends only 25 percent of its research and development expenditures on long-term research, down from 36 percent in 1957."[4]

On the other hand, high technology contributes substantially to our competitiveness in the international marketplace. The high-technology manufacturing industries in 1976 showed a favorable trade balance of $28 billion while other manufacturing industries showed a net deficit of $16 billion. Recent trends, however, suggest a leveling or even a weakening in high-technology contributions to the trade balance.[5]

Within the foreseeable future the costs of research, development and operations in space will remain relatively high. But the importance of involving the private sector early in space activities and with determination to make substantial strides will require major efforts in investigating appropriate institutional, financial and legal frameworks. With business activity must go a business climate where the private sector understands the rules and feels comfortable with their prospects for commercial success. American industries will be competing in the world marketplace where governmental policies and relationships with industry can vary substantially. During hearings of the House Subcommittee on Space Science and Applications, Sherwood Fawcett in describing overseas activities of Battelle Memorial Institute testified that:

> "In most advanced technical countries, industry and government are so closely related that it is sometimes difficult to distinguish between them."[6]

In the past, the American system of free enterprise has provided unparalleled national growth. In looking to private industrial participation in space, we cannot ignore the realities of present world and national economics. However, the apparent dichotomy of differing criteria in the United States for governmental and private sector expenditures need not preclude serving common interests but could contribute to less

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[4]U. S. Congress, House Committee on Science and Technology, Hearings on The Federal R & D Budget, April 5, 1979, 96th Cong., 1st Sess. at 227 (1979).

[5]United States Foreign Trade Annual, August 1979. OBA 79-22, Table 5 at 10-11, Table 6 at 14-15, published by the Office of International Economic Research of the U. S. Dept. of Commerce.

[6]Space Industrialization Act of 1979, Hearings Before the Subcommittee on Space Science and Applications, May 22, 23 and June 26, 27, 1979. (In print).

than full utilization of national resources and capabilities. Within the American tradition of free enterprise with minimum governmental interference, new approaches to high technology stimulation need to be thoroughly explored and evaluated.

Space Industrialization Corporation

One bold and innovative approach to addressing the needs of the private sector in advancing national capabilities in space is embodied in proposed legislation, H. R. 2337[7] The objective of this bill is to provide investment capital and establish the necessary business conditions to attract industries which can use the space environment to manufacture products and provide services.[8] The target industries are those which supply the goods and services to the private sector and not just the industries which provide aerospace equipment and systems to serve governmental purposes.[9]

The approach employed in H. R. 2337 is to provide a sufficient degree of flexibility to balance the interests of the nation and those of private enterprise in promoting aggressive and worthwhile space ventures. National interests would be served by ensuring that projects of the highest quality which advance the technology base of the nation and provide useful goods and services for domestic and world markets would be produced.[10] Private sector interests and objectives would be served by providing a source of investment capital which is committed to high-technology projects having significant prospect for commercial success but which otherwise can exceed acceptable risk to private shareholder investments.[11]

The mechanism of federal support seems particularly appropriate to space industrialization which has been nurtured from programs which have been essentially the exclusive domain of government. The complexity of technology and uncertainty in outcome were the obvious reasons for the national effort in opening up access to space and also provided impetus for a continued role of government in ensuring that the private sector can fully avail itself of new opportunities. Although earth orbit is shorter in distance than Washington, D.C., is from New York City, the private sector may well view this as an impenetrable abyss in the process from conceptualization to commercialization.

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[7]H. R. 2337, 96 Cong., 1st Sess., A Bill to Establish A Space Industrialization Corporation to Promote, Encourage and Assist in the Development of New Products, Processes, and Industries Using the Properties of the Space Environment. H. R. 2337 is essentially the same as H. R. 14297, 95 Cong., 2d Sess.

[8]*Id.*, §2(c).

[9]*Id.*, §102(a).

[10]*Id.*, §2(c) and (d).

[11]*Id.*, §102(9).

The major purpose of the legislation is to provide capital through direct equity investments, loans and loan guarantees. A trust fund would be established[12] and initially funded through federal appropriations.[13] The Secretary of the Treasury would administer the fund and the Space Industrialization Corporation would provide the institutional connection between the trust fund and private enterprise.[14] The Space Industrialization Corporation would conduct its operations in a manner familiar to the private sector and accordingly enter into arrangements with sound business basis.[15] The provisions of the bill recognize differences in business arrangements when private enterprise undertakes research and development for the competitive market as opposed to producing goods and services for government agencies. Typical contractor relationship with the government customer can often involve conditions which inhibit private enterprise commitments. In such cases, Government establishes the needs and specifies requirements, controls the funding and termination provisions and owns the property rights ensuing from the contract. The Space Industrialization Corporation in dealing with the private sector, on the other hand, would recognize commercial practices and make financial arrangements consistent with these needs while still protecting the interests of the taxpayer.

The Space Industrialization Corporation is essentially an investment bank having equity interests in space ventures. The Corporation is subject to the Government Corporation Control Act[16] to bring its activities under annual scrutiny by the Congress while providing operational flexibility to conduct a primarily business function as opposed to a governmental function. Several provisions of the bill are directed toward establishing a business-like relationship which would provide a high degree of assurance that a private entity could depend upon the continuity of obligated funds,[17] the commitment to management and financial plans,[18] security in handling competitive information,[19] private ownership of patent[20] and proprietary data[21] and the ultimate sharing in the benefits of the business venture.

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[12] *Id.*, §103(a).

[13] *Id.*, §103(b).

[14] *Id.*, §103(a).

[15] *Id.*, §102(d) and 101(f).

[16] 31 USC 856 (1945).

[17] H. R. 14297, 95th Cong., 2d Sess., §102(d)(3) and (f).

[18] *Id.*, §102(f).

[19] *Id.*, §102(l).

[20] *Id.*, §102(i).

[21] *Id.*, §102(i).

The extent to which Congress should accommodate these needs of private enterprise in promoting entry to space industrialization will require further consideration. Specifically, the Budget and Impoundment Control Act[22] exemptions on spending authority limitation would apply to the Corporation. The Freedom of Information Act[23] exemption for protection of commercial and proprietary information would be delineated as it applies to management and financial agreements to encourage high quality disclosures.[24] In addition, agreements entered into by the Space Industrialization Corporation can provide for full private ownership of intellectual property rights and subject all parties to performing the obligations contracted for.[25]

Since the objectives include both national purpose and business purpose, profitability is an important consideration in measuring the success of the Space Industrialization Corporation investments. Projects which are commercially successful would provide the cash flows necessary to reimburse the trust fund and form the basis for viable on-going operations.[26] The goal is that eventually federal appropriations would be returned to the U. S. Treasury[27] and a stockholder owned capital enterprise venture could be sustained to provide the source of private funds for infusion in space industrialization.[28] Less than ideal conditions may possibly result, however, and net profitability may never be achieved. In any event, national interests would still have been served if, on balance, the private sector had brought its efforts to bear in seeking opportunities for industrializing space. Commercially unsuccessful projects, although failures in a financial sense, could provide invaluable technical information for future projects.[29]

Summary

This article addresses a concept for financing private ventures which can have major effects on expanding uses of space for the benefit of the United States as well as for the benefit of all mankind. Questions of legal and policy significance are contained in

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[22]21 USC 1301 (1974).

[23]5 USC 552(b) (4) (1964).

[24]H. R. 14297, 95th Cong., 20 Sess., §102(l) (1978).

[25]*Id.*, §102(i).

[26]*Id.*, §102(f).

[27]31 USC 856 (1978), as amended by H. R. 14297 §106(a),
31 USC 867 (1978), as amended by H. R. 14297 §106(b),
31 USC 868 (1978), as amended by H. R. 14297 §106(c).

[28]H. R. 14297, 95th Cong., 2d Sess., §101(d).

[29]*Id.*, §102(g).

formative legislation to provide federal initiative for stimulating private sector innovation in order to enhance national capabilities in space. The programs of the National Aeronautics and Space Administration will continue the vital role of space exploration and technological development. The basic premise is that effective commercial applications require the commitment of private enterprise. These concepts as embodied by the legislation deserve consideration and discussion by thoughtful persons to define the proper role for government in addressing the risk and capital factors of future space activities.

Our experience in space exploration is just two decades old. The pace has been fast to be sure and the cement is still being poured. We know the ways in which space science has already expanded our horizon and we can only guess as to what the future will hold. The musings of men's minds are all interlocked and a single spark of light in one area can provide illumination to a vast succession of unforeseen connections to others.

What we know will lead us to the unknowns. What we discover there will bring us to the heretofore unimagined. This has been the history of civilization. Space industrialization is a stepping stone along this way—one on which we will balance for a while, gain a new perspective and continue on.